



University
of Idaho

ASSIGNMENT 4C: FINAL PRESENTATION

ANKIT PAUDEL

BIBEK SHARMA

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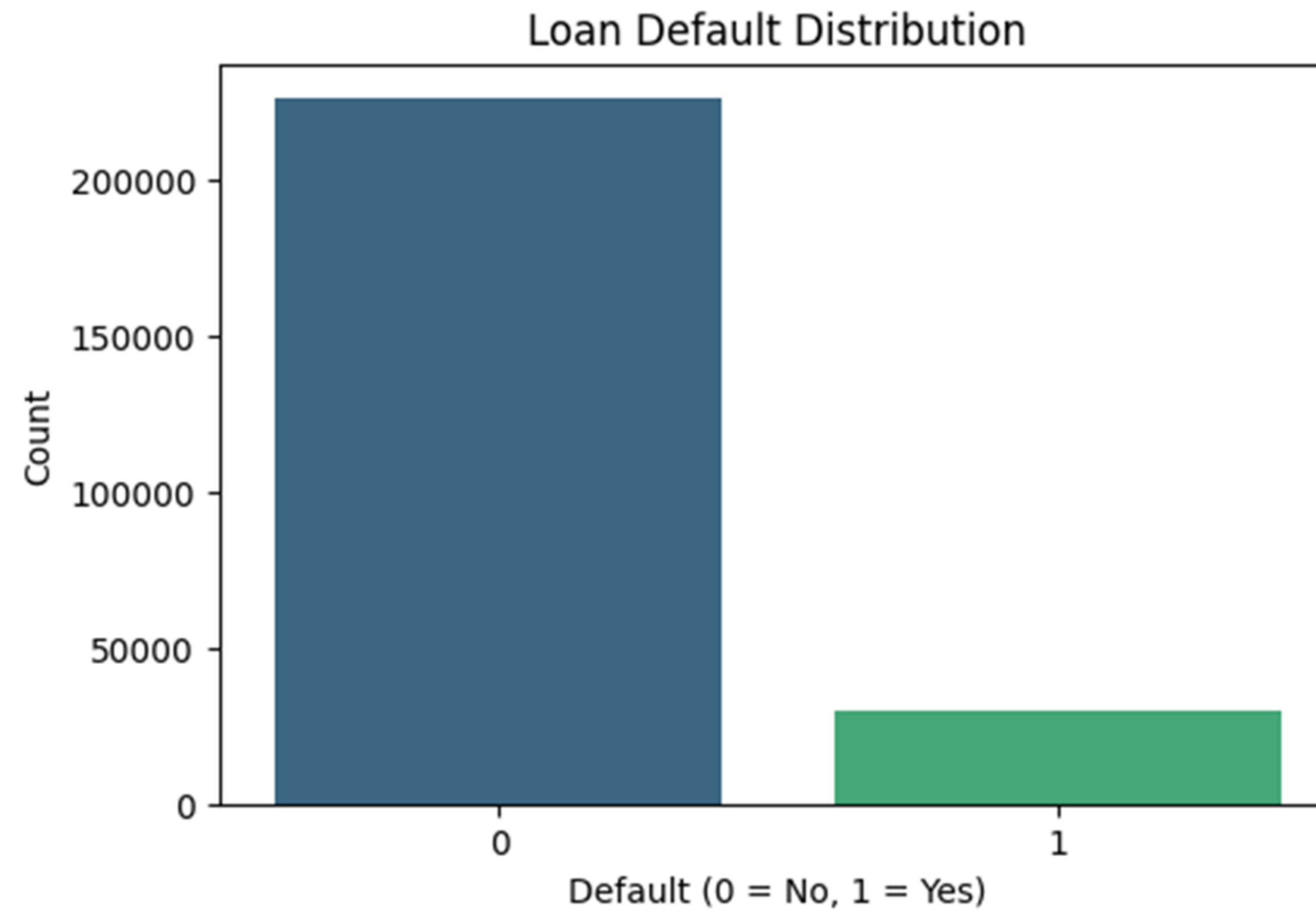
SOHAN LAMA

*** SUMIT SHAHI**

HYPOTHESIS QUESTIONS

- I Does a borrower's credit score have a significant impact on the likelihood of loan default, and can it be a primary predictor in the model?
- I Are income and loan amount jointly influential in predicting loan default, particularly when considering their interaction effects?

DATA DISTRIBUTION





METADATA

Loan Default Prediction Dataset Metadata

Dataset to predict which individuals will default on their loan payments

Dataset Overview

Creator: NIKHIL

Publisher: Kaggle

License: CC0: Public Domain

Date Modified: 2023-09-11

File: Loan_default.csv

Format: CSV

Size: ~7.858 MB

Records: 255,347

This dataset is from Coursera's Loan Default Prediction Challenge and contains information about borrowers to predict loan defaults.

Data Fields (18)

Field Name	Data Type	Description
LoanID	Text	A unique identifier for each loan
Age	Integer	The age of the borrower
Income	Integer	The annual income of the borrower
LoanAmount	Integer	The amount of money being borrowed
CreditScore	Integer	The credit score of the borrower
MonthsEmployed	Integer	The number of months the borrower has been employed
NumCreditLines	Integer	The number of credit lines the borrower has open
InterestRate	Float	The interest rate for the loan
LoanTerm	Integer	The term length of the loan in months
DTIRatio	Float	The Debt-to-Income ratio
Education	Text	The highest level of education attained by the borrower
EmploymentType	Text	The type of employment status of the borrower
MaritalStatus	Text	The marital status of the borrower
HasMortgage	Boolean	Whether the borrower has a mortgage
HasDependents	Boolean	Whether the borrower has dependents
LoanPurpose	Text	The purpose of the loan
HasCoSigner	Boolean	Whether the loan has a co-signer
Default	Integer	Indicate whether the loan defaulted or not

File Distribution

archive.zip (7.858 MB)

Archive containing all dataset contents

Loan_default.csv

Contains features for borrowers and loan default status

Keywords & Categories

Banking

Finance

Income

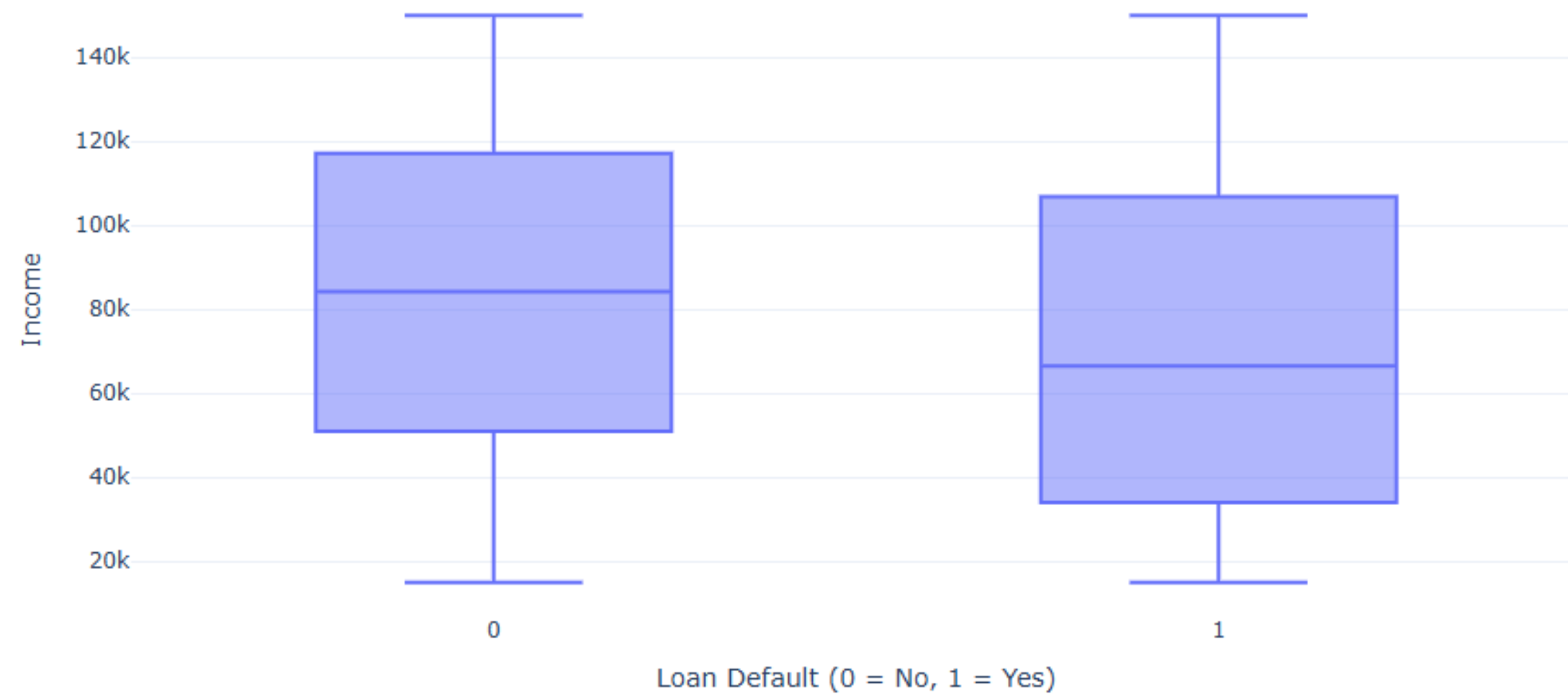
Binary Classification

This visualization represents the metadata structure for the Loan Default Prediction Dataset. To work with the actual data, you'll need to access the CSV file containing the 255,347 records.

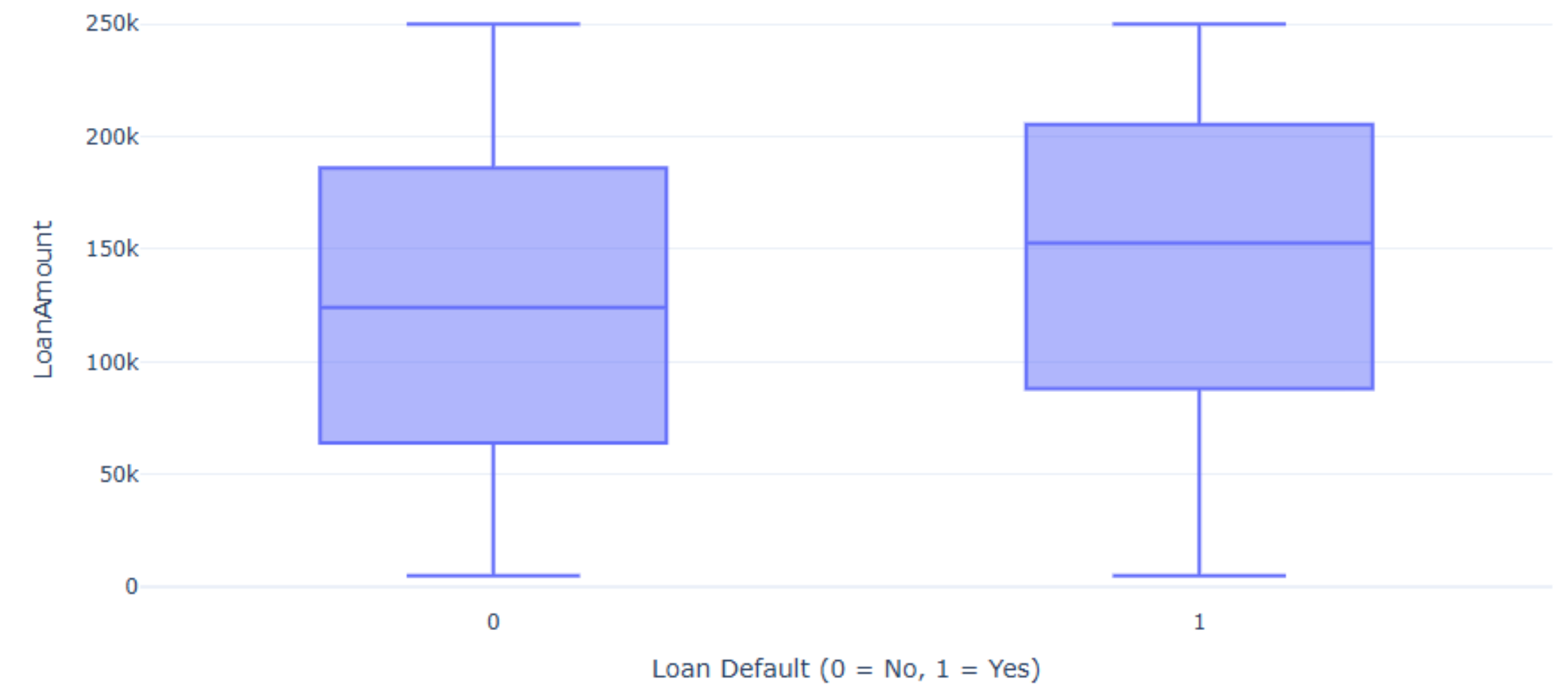
INITIAL DATA PRE-PROCESSING AND EDA



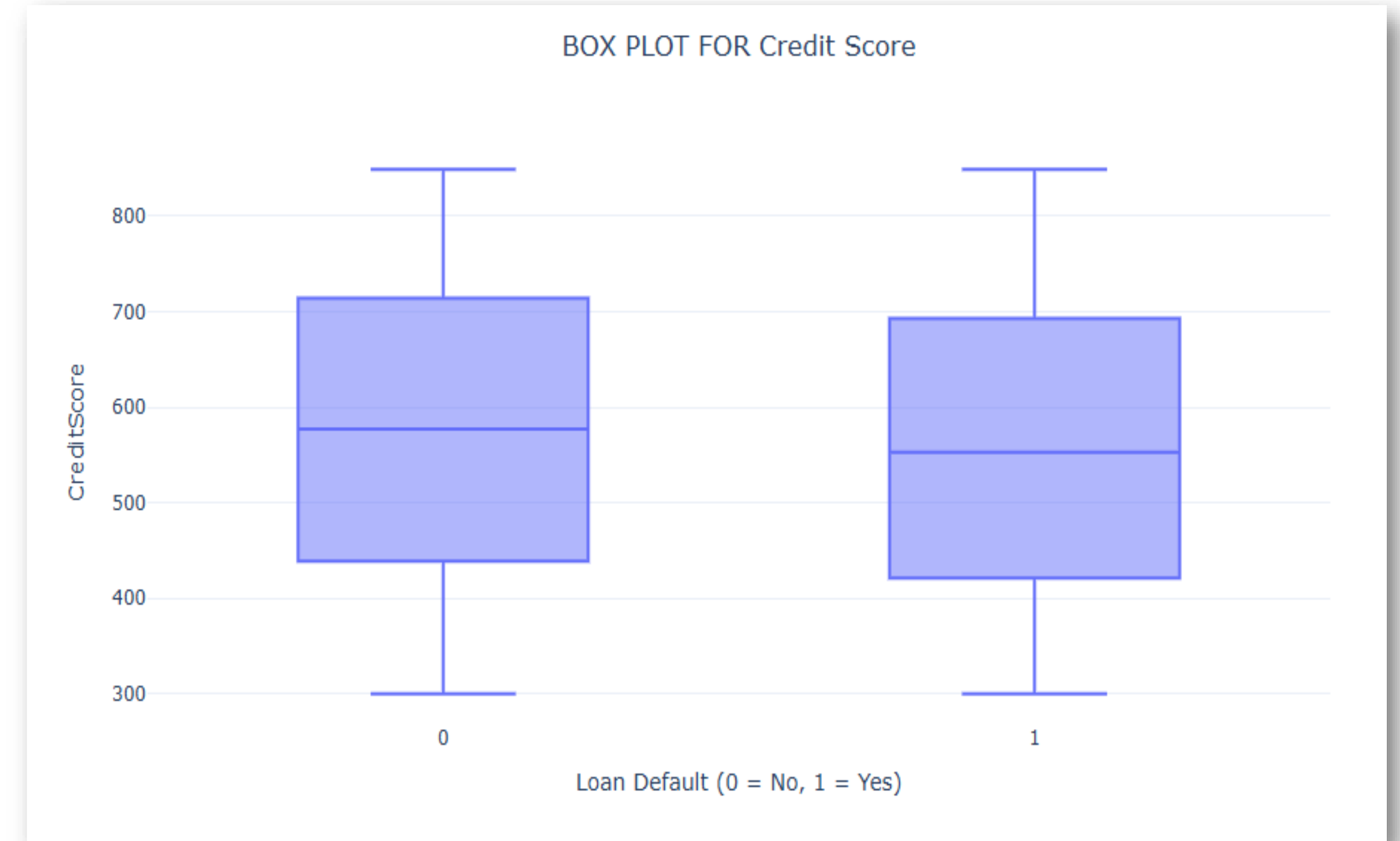
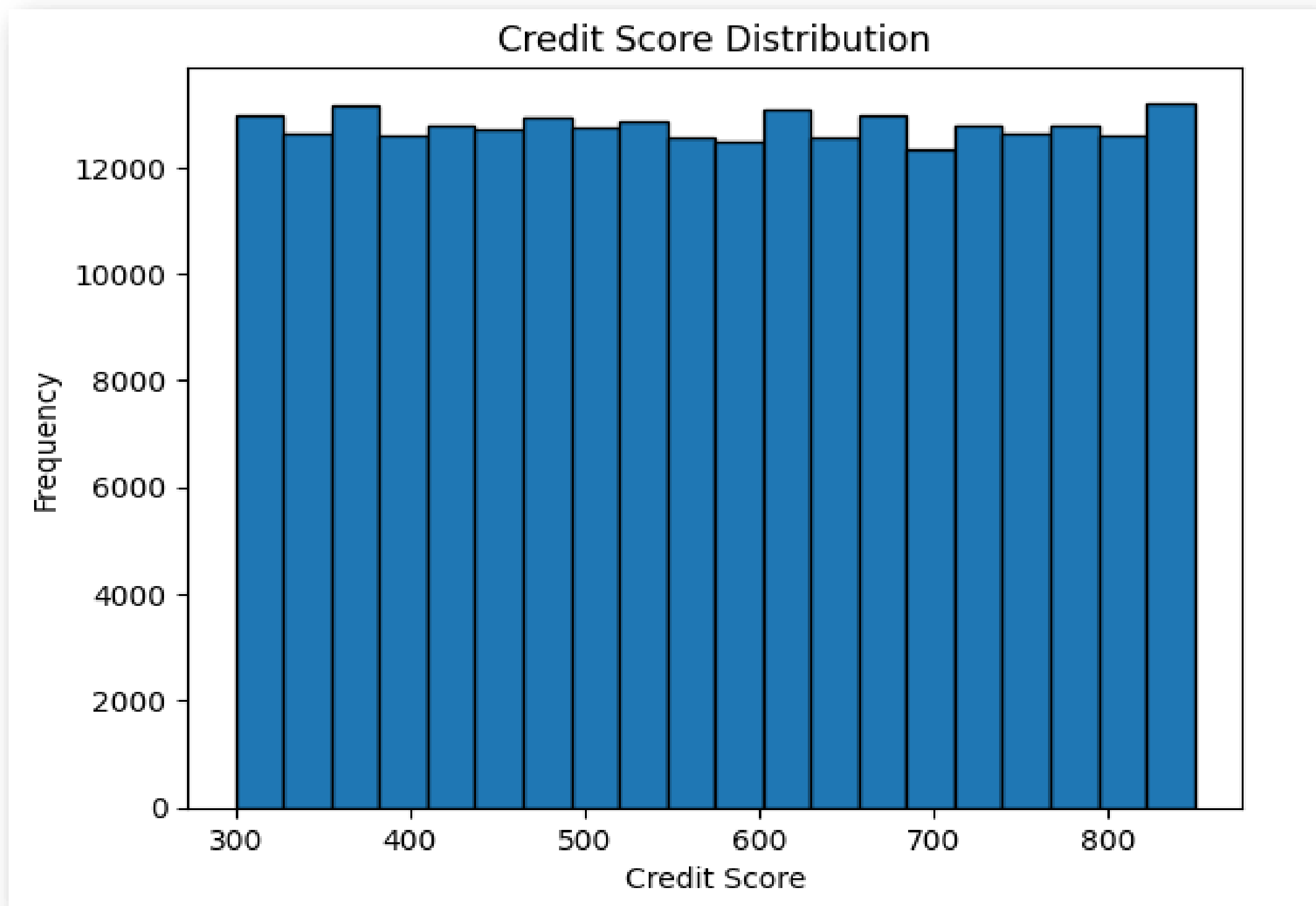
BOX PLOT FOR Income



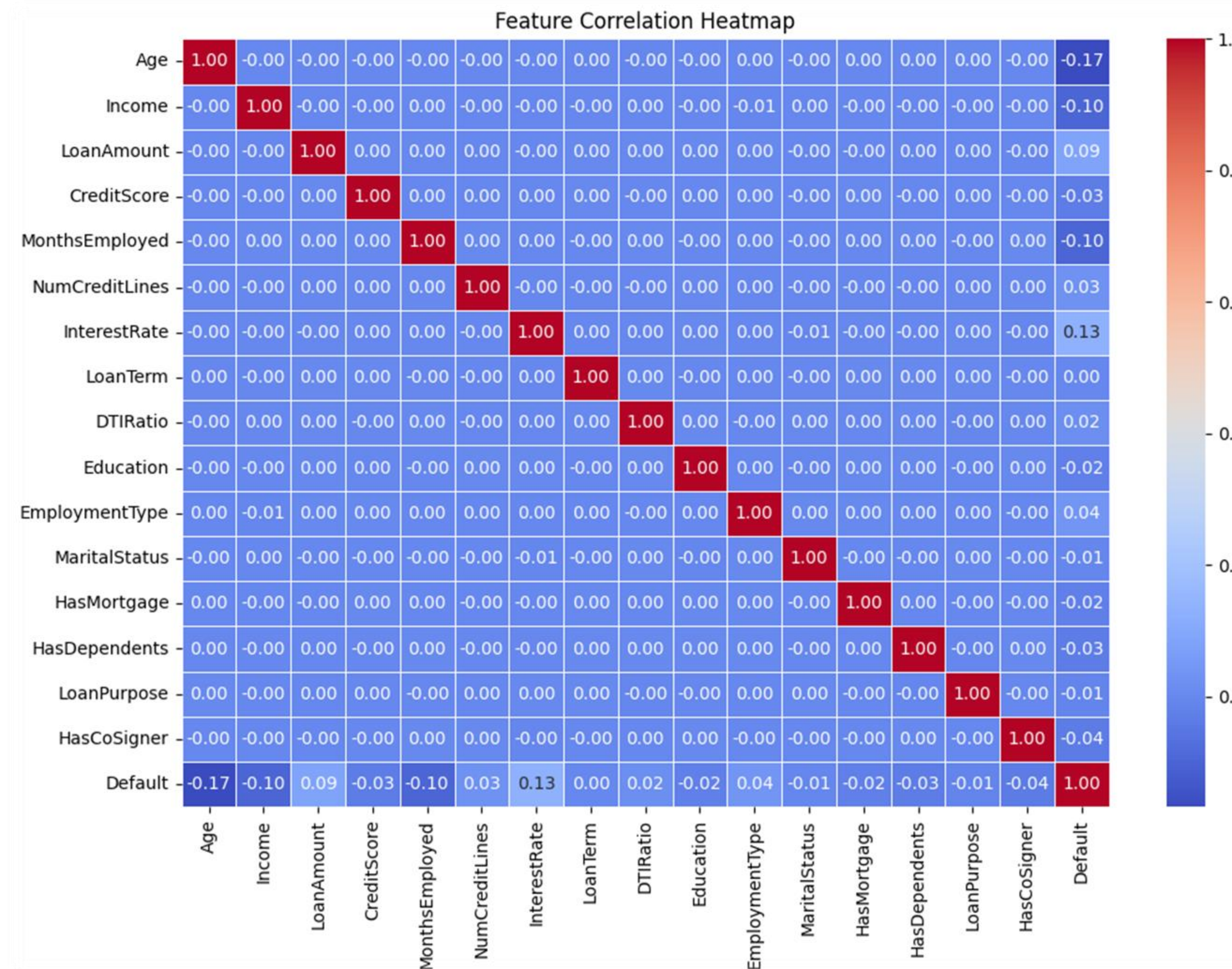
BOX PLOT FOR LoanAmount



INITIAL DATA PRE-PROCESSING AND EDA



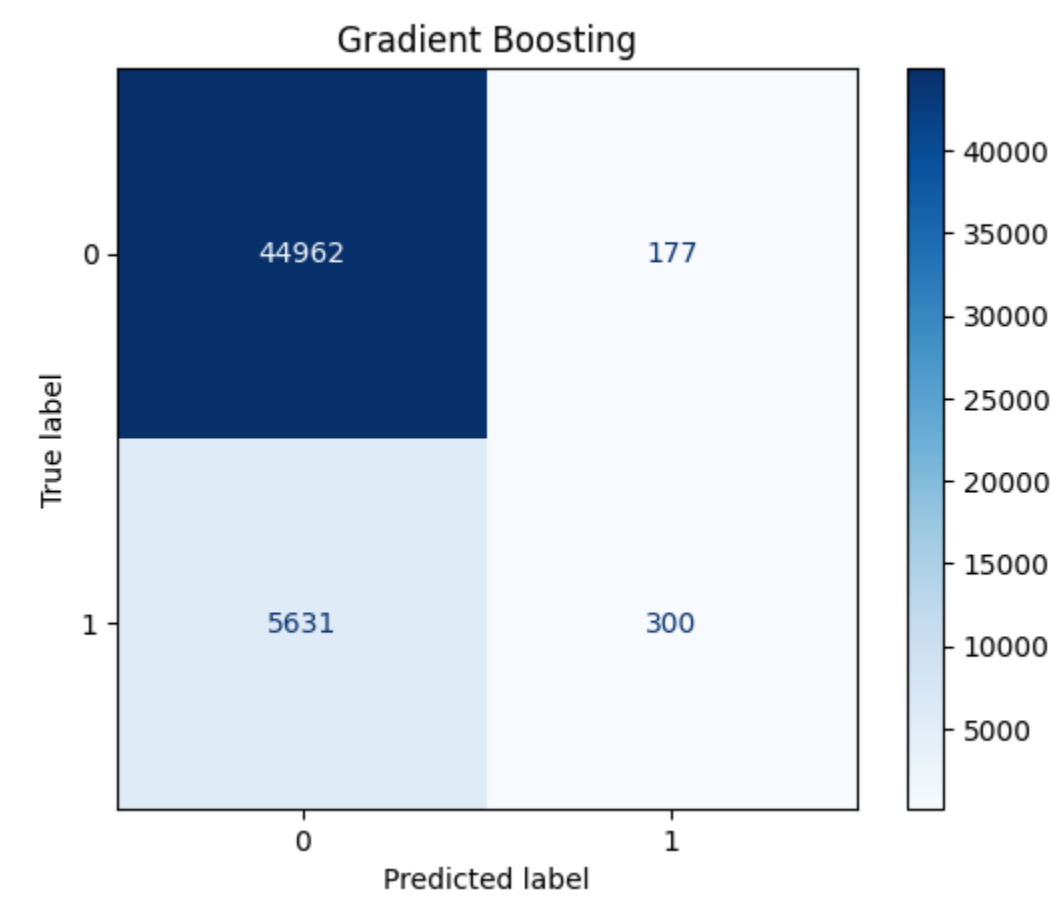
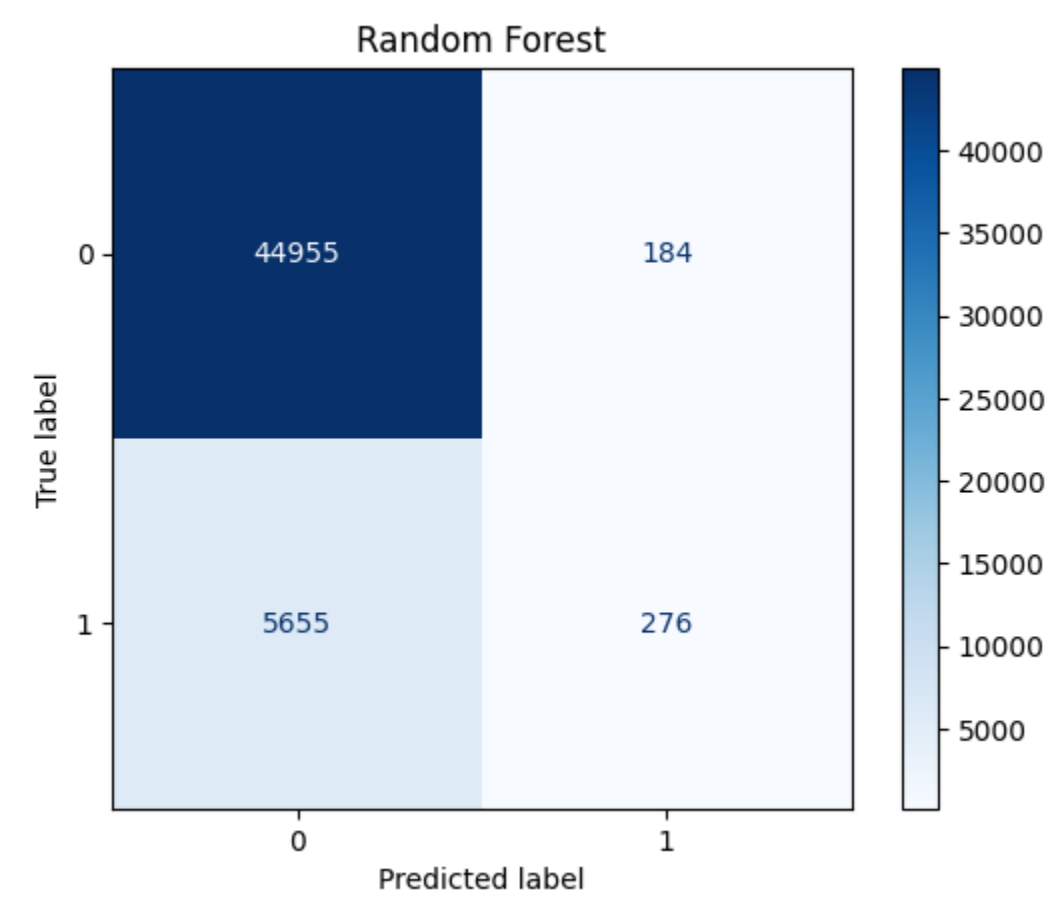
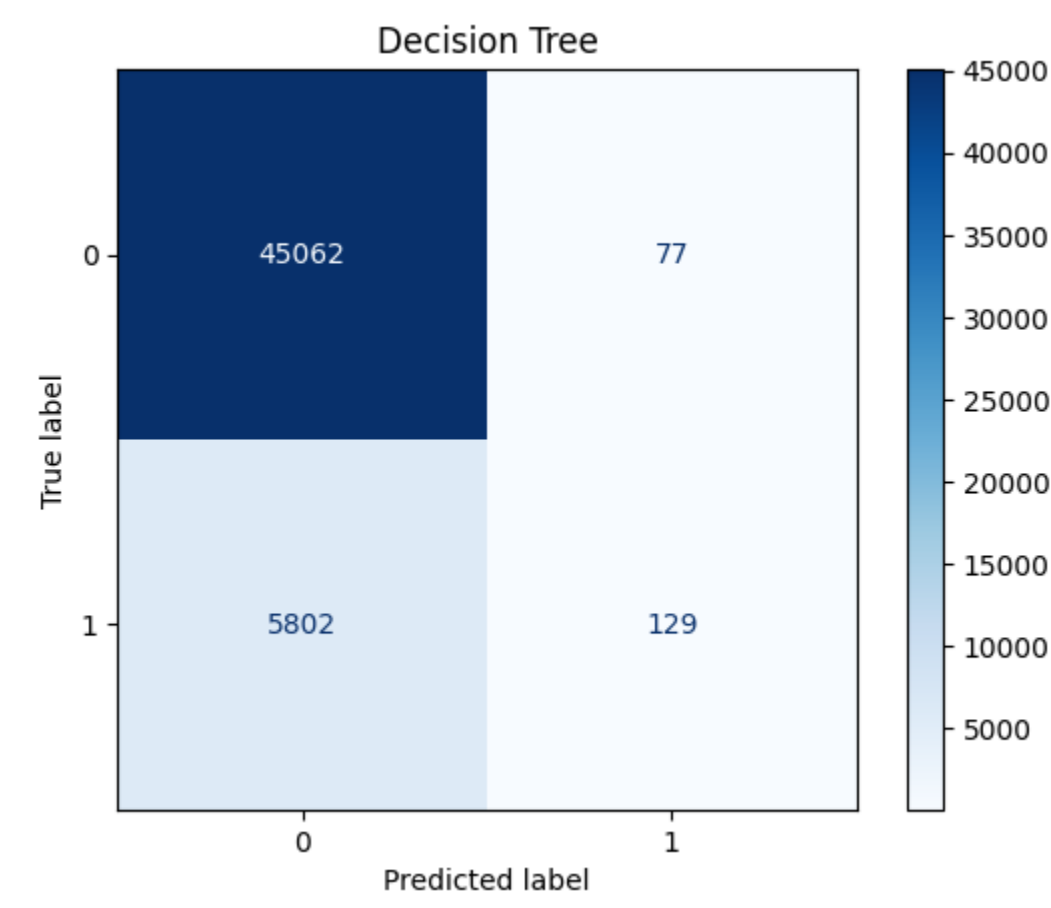
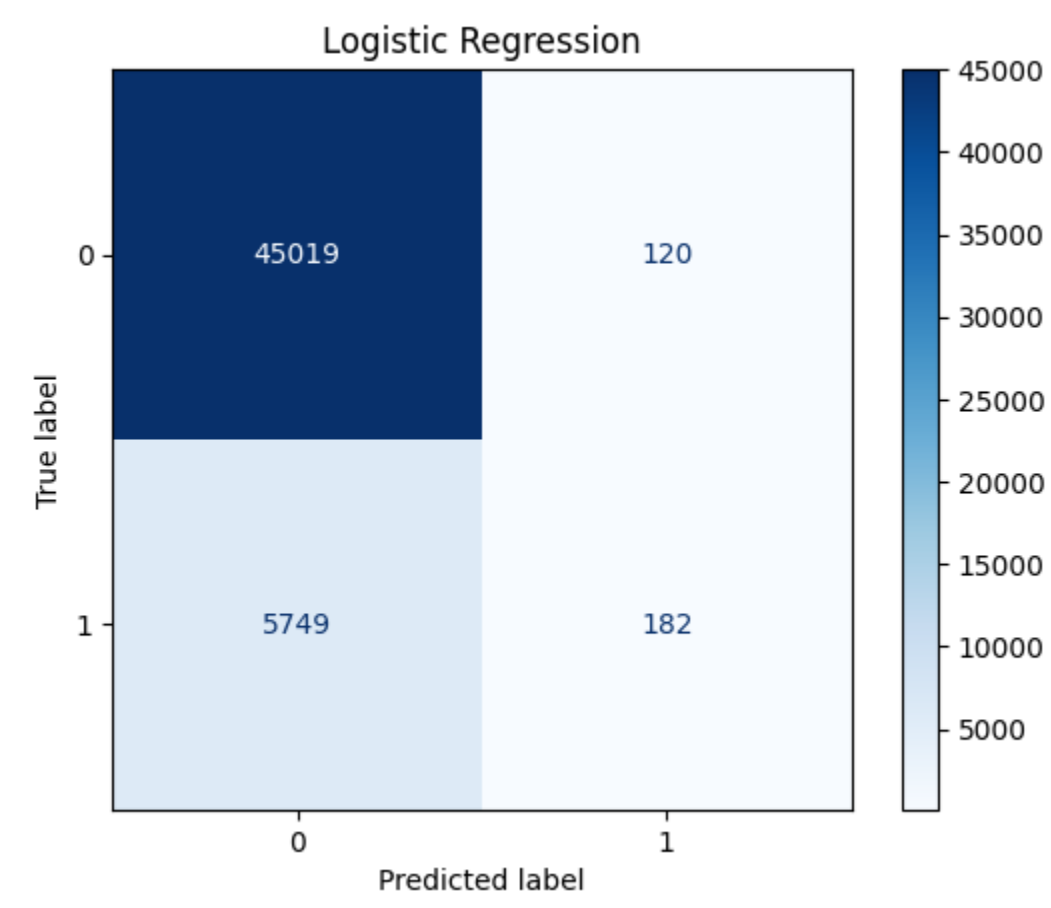
CORRELATION HEATMAP



ML MODEL ACCURACY COMPARISON



CONFUSION MATRIX



CONCLUSION



Results suggest our hypothesis questions may not fully hold.

I Possible reasons:

- Features are highly co-dependent.
- Significant class imbalance in the dataset.

I Recommendation for Future Work

- Use a dataset with more loan default examples.
- Apply techniques to address class imbalance.
- Explore methods to better isolate feature impacts.

REFERENCES:

I *Data Science Coding Challenge: Loan Default Prediction.* Coursera. (n.d.).
<https://www.coursera.org/projects/data-science-coding-challenge-loan-default-prediction>

I Nikhil. (2023, September 11). *Loan default prediction dataset.* Kaggle.
<https://www.kaggle.com/datasets/nikhil1e9/loan-default/data>

ANY
QUESTIONS



Thank You